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**One Fundamental and Two Taxes: When Does a Tobin Tax Reduce Financial
Price Volatility?**

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1 Big picture

- **Question.** Does a securities transaction tax, or Tobin tax, reduce financial price volatility? If the answer depends on market maturity, when does the sign flip?
- **Research setting.** The paper studies Chinese firms cross-listed in Mainland China and Hong Kong. Mainland A-shares and Hong Kong H-shares have identical corporate fundamentals but face different stamp duties and, because of capital controls, are traded in segmented markets.
- **Treatment-control logic.** A-share stamp duties change repeatedly, while H-share stamp duties are low and nearly unchanged. H-shares therefore provide a close control group for the same firms' fundamentals.
- **Average result.** Pooling seven A-share stamp-duty changes, higher transaction costs are associated with lower relative A-share volatility. This supports the Tobin-tax intuition in a market dominated by retail/noise traders.
- **Heterogeneity result.** As institutional investors become important, the effect reverses. With high institutional trading/ownership, higher transaction costs are associated with higher volatility.
- **Mechanism.** A tax can reduce volatility if it deters non-fundamental trading more than informed trading. But in a more developed market, a tax may deter informed fundamental traders and slow price discovery, raising volatility.
- **Contribution.** The paper reconciles mixed empirical results by arguing that the effect of a transaction tax is not universal; it depends on who trades.
- **Policy takeaway.** A Tobin tax may work in immature retail-dominated markets but can backfire in developed institution-dominated markets.

2 Incremental contribution of the paper

1. **Cleaner counterfactual.** The paper improves on standard before-after and double-difference designs by using the same firms' H-shares as controls for A-shares. This creates one fundamental cash-flow stream and two transaction-tax regimes.
2. **Two-market identification with capital controls.** Capital controls limit arbitrage between A and H markets, so tax differences can affect trading and volatility in one market without immediately being arbitrated away.
3. **Context-dependent Tobin-tax effect.** The paper's main conceptual innovation is that the sign of the tax-volatility relationship depends on market composition. Retail/noise-trader dominated markets and institution-rich markets respond differently.
4. **Reconciliation of contradictory literature.** Prior empirical studies disagree on whether transaction taxes reduce volatility. This paper offers a unifying explanation: earlier studies are estimating different regimes.
5. **Dynamic market-development perspective.** China's rapid transition from a frontier retail market to a market with large institutional participation allows the authors to observe multiple market regimes in one institutional setting.
6. **Firm-level and event-level evidence.** The paper combines firm-event regressions, portfolio-level regressions, placebo tests, event-level plots, and institutional interaction/IV tests.
7. **Policy discipline.** The study highlights that transaction taxes should not be evaluated as a generic tool; their effectiveness depends on the relative prevalence of informed and uninformed trading.

3 Data and measurement

3.1 Cross-listed firms and sample

The sample consists of Chinese firms simultaneously listed as Mainland A-shares and Hong Kong H-shares. A-shares and H-shares have the same underlying corporate fundamentals and cash-flow rights, but are traded by different investor pools and face different stamp duties.

- **Treatment group:** Mainland A-shares, directly affected by Mainland stamp-duty changes.
- **Control group:** Hong Kong H-shares of the same firms, with nearly constant and much lower stamp-duty changes.
- **Sample size:** 223 unique firm-event observations across seven A-share stamp-duty events.
- **Data sources:** CSMAR, Wind, Reuters Datastream, and Pacific Basin Capital Market data.

Interpretation: The key advantage is that A and H shares of the same firm have identical fundamentals, making H-shares a far better counterfactual than unrelated control stocks.

3.2 Stamp-duty events

The paper studies seven Mainland A-share stamp-duty adjustments from 1997 to 2008:

- May 1997: tax increase;
- June 1998, November 2001, January 2005: tax reductions;
- May 2007: tax increase;
- April 2008 and September 2008: tax reductions.

Interpretation: These changes supply quasi-events in transaction costs. The 2008 reductions are especially important because institutional investors were far more important by then.

3.3 Main outcomes

The paper measures A-share volatility and H-share volatility around each event, generally using 240 trading days before and after a stamp-duty change. It defines relative volatility outcomes such as:

$$\Delta\sigma_{m,i,t} = \sigma_{m,i,t}^A - \sigma_{m,i,t}^H, \quad (1)$$

where m indexes events, i indexes firms, and t indexes the high-tax or low-tax regime.

Alternative outcomes include:

- log volatility ratios $\log(\sigma^A/\sigma^H)$;
- variance differences $V(A) - V(H)$;
- turnover ratios $\log(T(A)/T(H))$;
- price reactions around announcement days.

Interpretation: The relative outcome subtracts the H-share behavior of the same firm, removing fundamental shocks common to both share classes.

3.4 Institutional investor measures

Institutional activity is measured using:

- **Inst_own:** institutional ownership as a fraction of tradable shares;
- **Inst_turn:** institutional turnover, measured from changes in institutional holdings;
- **Inst_size:** number of institutional investors in the stock.

Interpretation: The institutional variables proxy for the presence of more fundamental-based traders. They are crucial for the paper’s claim that the Tobin-tax effect is regime dependent.

4 Model intuition and empirical specifications

4.1 Price decomposition intuition

The paper models A-share and H-share returns as containing a common fundamental component and market-specific speculative components. A simplified return decomposition is:

$$r_{it}^A = d_{it} + k_{it}^A + s_{it}^A, \quad r_{it}^H = d_{it} + s_{it}^H, \quad (2)$$

where d_{it} is fundamental news, k_{it}^A captures discount-rate changes in the A-share market, and s_{it}^A, s_{it}^H are speculative components.

Under the identifying logic, changes in relative volatility around a Mainland tax change trace changes in the speculative component of A-shares relative to H-shares:

$$\frac{\Delta \text{Var}(r^A) - \Delta \text{Var}(r^H)}{\Delta \text{tax}^A} \approx \frac{\Delta \text{Var}(s^A)}{\Delta \text{tax}^A}. \quad (3)$$

Interpretation: The point of the A-H comparison is to remove d_{it} , the fundamental component. What remains is the part of price variation plausibly affected by trading frictions and speculative/noise trading.

4.2 Baseline volatility regression

The main firm-level specification is:

$$\Delta \sigma_{m,i,t} = \gamma_0 + \gamma_1 \text{HighTax}_{m,t}^A + C_{m,t} + F + \varepsilon_{m,i,t}. \quad (4)$$

- $\Delta \sigma_{m,i,t}$ is A-share volatility minus H-share volatility.
- $\text{HighTax}_{m,t}^A$ equals one in the higher-stamp-duty regime.
- $C_{m,t}$ captures event-regime controls such as interest rates and exchange rates.
- F includes firm and event fixed effects in richer specifications.

Interpretation: A negative γ_1 means that higher A-share stamp duties reduce A-share volatility relative to the same firm’s H-share volatility.

4.3 Heterogeneity by institutional participation

The core heterogeneity regression is:

$$\Delta \sigma_{m,i,t} = \gamma_0 + \gamma_1 \text{HighTax}_{m,t}^A + \gamma_2 \text{IO}_{m,i} + \gamma_3 \text{HighTax}_{m,t}^A \times \text{IO}_{m,i} + C_{m,t} + F + \varepsilon_{m,i,t}. \quad (5)$$

Interpretation: The coefficient γ_3 is central. If it is positive, the volatility-reducing effect of a tax becomes weaker, or reverses, when institutional trading is high.

4.4 Endogeneity and placebo tests

The paper tests whether stamp-duty changes are predictable from market returns or volatility and performs placebo regressions using fake event dates predicted from non-event periods.

Interpretation: These tests address the concern that the government changes stamp duties in response to volatility. If fake events do not produce significant coefficients, the baseline results are less likely to be spurious.

5 Identification and empirical design

5.1 Core identification

The design combines three ingredients:

1. the same firm has A and H shares, giving one fundamental cash-flow process;
2. capital controls segment the two markets, limiting arbitrage;
3. A-share and H-share stamp duties change at different times and magnitudes.

Interpretation: This creates a cleaner treatment/control comparison than comparing unrelated exchanges or different firms.

5.2 Why H-shares are a strong control

H-shares have the same issuer, same profits, and same corporate shocks as A-shares. If a company receives good or bad fundamental news, both share classes should be affected. Differences in relative volatility around A-share tax changes can therefore be attributed more plausibly to market-specific trading conditions.

Interpretation: The H-share control does not make the design perfect, because markets can still differ in investor composition and liquidity, but it removes the most important confound: firm fundamentals.

5.3 Threats and responses

- **Policy endogeneity.** The government may adjust taxes in response to market conditions. Table 5 and placebo Table 9 address this.
- **Liquidity differences.** A and H shares differ in liquidity and float. Table 8 adds liquidity and float controls.
- **Short-sale constraints.** H-shares can sometimes be shorted while A-shares cannot. Table 7 and related robustness tests address alternative volatility measures.
- **Changing market composition.** Institutional participation rises over time. Tables 13-16 and Figures 4-7 explicitly model this changing composition.

Interpretation: The identification is strongest when the average effect, robustness checks, and institutional heterogeneity all tell a coherent story.

5.4 Interpreting the sign

A negative effect of *HighTax* on volatility suggests the tax deters noise or speculative trading more than informed trading. A positive effect suggests the tax deters informed fundamental trading enough to reduce price discovery and raise volatility.

Economic message: This is the paper’s policy message: the sign of the Tobin-tax effect is market-state dependent.

6 Tables and figures

6.1 Table 1: Historical adjustment of stamp duties; Figure 1: Evolution of stamp duties

Read: Table 1 lists all A-share and H-share stamp-duty changes after 1993. Figure 1 visualizes the much larger and more frequent A-share changes compared with the nearly flat H-share duty.

Economic message: The table and figure define the policy variation. A-shares experience large tax shocks; H-shares provide the low-change control.

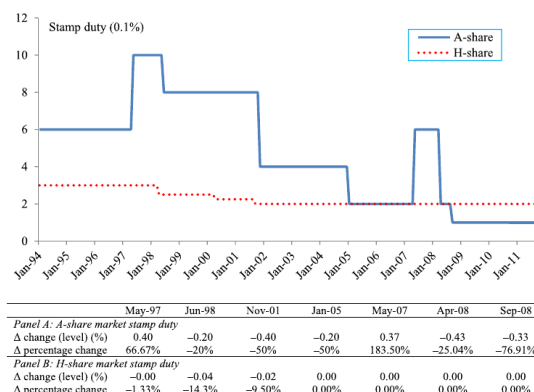
Interpretation: The small H-share adjustments add noise rather than driving the estimates, which biases against finding clean effects.

Table 1

Historical adjustment of stamp duties on stock trading in China and Hong Kong. Lists all the historical adjustments of stock trading stamp duty in China (A-share) and Hong Kong (H-share) markets after 1993 when the first cross-listed firm appeared. Unless specified, otherwise stamp duty reported below aggregates the tax rate of both buyer and seller sides.

Date	Activities
<i>Panel A: historical adjustments of stamp duty on stock trading in China</i>	
12-May-97	Trading stamp tax increased from 0.6% to 1.0%
12-Jun-98	Trading stamp tax reduced from 1.0% to 0.8%
16-Nov-01	Trading stamp tax reduced from 0.8% to 0.4%
23-Jan-05	Trading stamp tax increased from 0.4% to 0.2%
30-May-07	Trading stamp tax increased from 0.2% to 0.6%
24-Apr-08	Trading stamp tax reduced from 0.6% to 0.2%
19-Sep-08	Trading stamp tax for buyer abolished, while stamp tax for seller remains at 0.1%
<i>Panel B: historical adjustments of stamp duty on stock trading in Hong Kong</i>	
1-Apr-98	Trading stamp tax reduced from 0.3% to 0.25%
7-Apr-00	Trading stamp tax reduced from 0.25% to 0.225%
1-Sep-01	Trading stamp tax reduced from 0.225% to 0.2%

Table 1: historical A-share and H-share stamp-duty events.



on of stamp duties in China and Hong Kong. The evolution of trading stamp duty (sum over buyers and sellers) in A-share and H-share markets from January 1994. The level of stamp duty to both buyer and seller in a unit of 0.1%. The table below shows the average level and percentage change of stamp duty around each A-share stamp duty change. Δ change (level) is calculated as average monthly stamp duty over 12 months when the A-share stamp duty changes minus the average monthly stamp duty over the 12 months on and after the month when the A-share stamp duty changes.

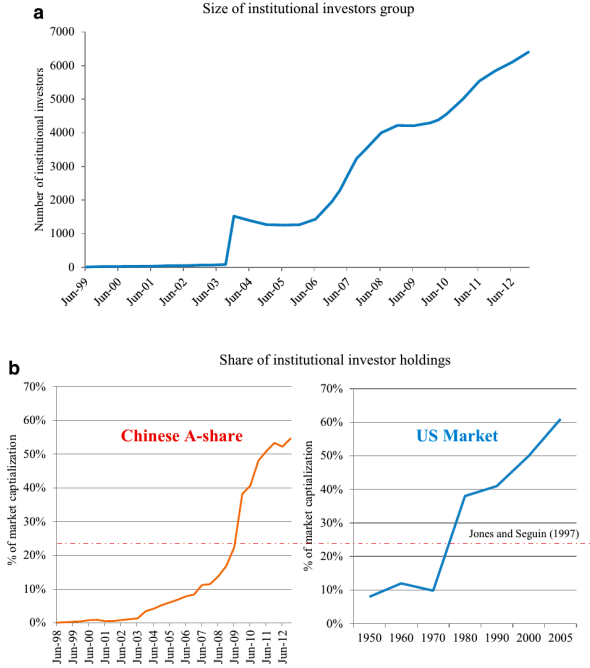
Figure 1: time-series evolution of stamp-duty levels.

6.2 Figure 2: Evolution of institutional investors; Table 2: Summary statistics

Read: Figure 2 documents the dramatic growth of institutional investors in China. Table 2 reports firm and stock characteristics for cross-listed stocks and the broader A-share market.

Economic message: The figure motivates the paper’s regime-switching story; Table 2 shows that cross-listed firms are large, liquid, and increasingly institutionally held.

Interpretation: China’s market moves from retail dominance to substantial institutional participation within the sample, allowing the paper to test whether the Tobin-tax effect changes as the trader mix changes.



g. 2. Evolution of institutional investors in China. Information on Chinese institutional holding is from the Wind database on semi-annual basis from 1999 to 2012. (a) reports the absolute number of institutional investors in the Chinese market. The left and right panels of (b) report the proportion of institutional holding with respect to the marketization of tradable shares in the Chinese A-share market (1999-2012) and the US market (1959-2006), respectively. The horizontal line indicates the proportion of institutional holding in the US market.

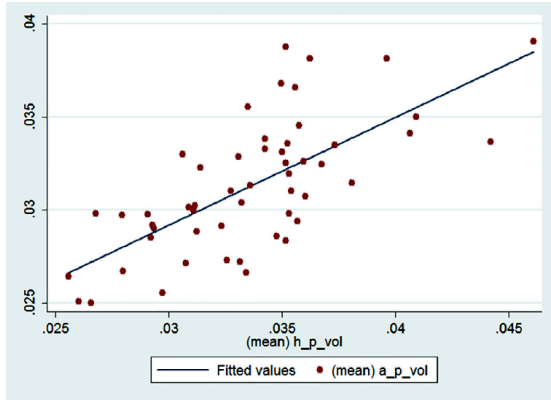
Figure 2: institutional investor growth in China.

6.3 Figure 3: Scatter plots of price volatility; Table 3: Variable correlation matrix

Read: Figure 3 shows a positive relationship between average A-share and H-share volatility across cross-listed firms. Table 3 reports correlations among volatility, returns, turnover, institutional variables, stamp-duty changes, interest rates, and exchange rates.

Economic message: The positive A-H volatility correlation supports the idea that H-share volatility is an informative control for A-share volatility.

Interpretation: The correlation is far from perfect, which leaves room for market-specific speculative or liquidity components affected by taxes.



plots of price volatility in the two markets. The x-axis is the H-share price volatility averaged over the sample period, and the y-axis is the A-share price volatility averaged over the sample period.

Figure 3: A-share versus H-share price volatility.

Table 2 Summary statistics. The sample contains 223 unique firm-event observations from 1997 to 2008. In Panel A, we first provide the mean and median of firm characteristics for our sample stocks and the entire A-share market in 1996 and 2008, respectively. The information is acquired from CSMAR and Wind for the annual filing (December). Inst_own is calculated instructional ownership as a fraction of total number of tradable shares in the reporting period. Inst_turn is measured as the absolute change in the institutional ownership with respect to one quarter prior in the reporting period. Inst_size is total number of institutional investors for each quarter and obtained directly from Wind database. In Panel B, stock characteristics of sample A- and H-shares are reported, respectively. For each variable, it takes the average of the daily value over a 480-trading-day period around each stamp duty change. All the value variables are reported in RMB, and for H-shares, we convert the HKD to RMB based on the historical daily exchange rate. H-share return volatility considers volatilities both of daily H-share return and of daily exchange rate between HKD and RMB. Appendix 1 has the details on variable definition.

Panel A: firm characteristics				
Currency: RMB	1996		2008	
	Cross-listed sample	A-share market	Cross-listed sample	A-share market
	Mean/median	Mean/median	Mean/median	Mean/median
No. observations	13	602	31	1712
TA (mil)	5989/3244	1244/459	703,067/56,459	9427/1959
Sales (mil)	3041/1516	632/301	108,805/36,836	4586/1191
EBIT/sales	7.3%/5.4%	14.1%/10.7%	11.5%/5.5%	-1.6%/4.3%
NI/sales	5.9%/3.9%	12.6%/10.1%	9.2%/4.3%	2.9%/4.7%
Sales growth	-2.9%/1.2%	8.1%/2.3%	15.7%/17.5%	17.8%/10.0%
Cash/TA	16.7%/19.1%	10.0%/8.0%	11.5%/8.1%	16.0%/12.8%
Debt/TA	6.0%/3.9%	5.1%/2.2%	11.9%/8.6%	5.9%/1.0%
Inst_own	0.0%/0.0%	0.0%/0.0%	35.1%/29.3%	23.2%/17.3%
Inst_turn	0.0%/0.0%	0.0%/0.0%	15.6%/9.2%	7.7%/2.7%
Inst_size	0/0	0/0	62/31	16/6
Panel B: stock characteristics				
Currency: RMB	A-share		H-share	
	Mean/median		Mean/median	
Mkt_Cap (mil)	129,292		32,612	
P/B	13.7		6.4	
Volume (mil)	282		252	
R	0.02%		0.01% (-0.01%)	
V	0.0330		0.0380	
T	0.0055		0.0115	
Illiqu	0.0014		0.0325	

^a Daily return of H-share calculated with stock price in RMB is reported in parentheses. It is calculated as $r_{i,t}^{FX} = (r_{i,t}^H + 1) \times (r_{t,t}^{FX} + 1) - 1$.

Table 2: sample and market summary statistics.

Table 3 Variable correlation matrix.

Provides the correlation matrix of the key variables pooled over seven events of stamp duty changes and 480 days around each event. First, for volatility, return and turnover, we calculate or average over a window of 480 trading days around (240 trading days before and 240 trading days after) each stamp duty change. In particular, V(A) and V(H) are volatility of A- and H-shares calculated as standard deviation of daily return over 480 trading days around each stamp duty change. R(A) and R(H) refer to average daily stock return, and T(A) and T(H) refer to the average daily share turnover over the same window. R(H) and V(H) also consider the effect of exchange rate change. Second, other variables are calculated using values prior to each stamp duty change. Inst_own refers to the institutional ownership obtained in the latest quarter prior to each stamp duty change. For the sample A-shares, Inst_turn refers to the summation of the absolute value of changes in ownership of each institutional investor in the latest quarter prior to stamp duty changes relative to one quarter before. Inst_size is the number of institutional investors for each sample firm in the latest quarter prior to each stamp duty change. L_STT is the initial level of stamp duty in the A-share market prior to each stamp duty change, while ΔSTT refers to the change in stamp duty for each stamp duty change. Lastly, R_{FX} in China and HK are monthly rates reflecting the 12-month base rates average over 480 trading days around each stamp duty change, while FX refers to the average value of HKD/RMB over the same window. Details on the variable definition can be found in Appendix 1.

	V(A)	V(H)	R(A)	R(H)	T(A)	T(H)	Inst_own	Inst_turn	Inst_size	L_STT	ΔSTT	R _{FX}	R _{FX}	FX
V(A)	1													
V(H)	0.53	1												
R(A)	0.31	-0.09	1											
R(H)	-0.14	-0.36	0.50	1										
T(A)	0.61	0.23	0.29	-0.10	1									
T(H)	0.23	0.36	0.13	0.05	0.09	1								
Inst_own	0.15	-0.15	-0.02	0.17	-0.03	0.18	1							
Inst_turn	0.12	-0.11	0.02	0.16	0.02	0.20	0.73	1						
Inst_size	0.03	-0.10	-0.09	0.09	-0.05	0.18	0.63	0.62	1					
L_STT	-0.33	0.35	-0.26	-0.27	-0.30	-0.04	-0.37	-0.39	-0.36	1				
ΔSTT	0.20	-0.14	0.61	0.21	0.16	0.02	-0.11	0.00	-0.04	-0.55	1			
R _{FX}	0.01	0.38	0.05	-0.43	-0.11	-0.03	-0.31	-0.27	-0.21	0.40	0.33	1		
R _{FX}	-0.12	0.03	0.55	0.11	-0.08	0.00	-0.43	-0.35	-0.43	0.32	0.56	0.55	1	
FX	0.61	0.20	-0.09	-0.11	0.38	0.12	0.41	0.33	0.36	-0.51	-0.13	-0.43	-0.66	1

Table 3: key variable correlations.

6.4 Table 4: Price response to stamp-duty changes; Table 5: Determinants of stamp-duty changes

Read: Table 4 shows that stamp-duty increases are associated with lower event-window returns. Table 5 tests whether changes in stamp duty are predicted by prior returns or volatility.

Economic message: Table 4 describes price reactions; Table 5 addresses policy endogeneity by asking whether tax changes are predictable from prior market behavior.

Interpretation: The fact that return lags predict some changes but relative volatility lags are weak supports the view that tax changes are not simply volatility-targeting events.

Table 4

Price response to stamp duty changes.

This table reports the price reaction of sample shares to the announcement of stamp duty changes. In Columns 1 and 2, we follow the specification as $r_{m,i}^A - r_{m,i}^H = \beta_0 + \beta_1 \text{TaxHike}_m^A + C_m + \varepsilon_m$, where the dependent variable is the portfolio return difference between A-shares and H-shares over the [+2 days,+1 day] window around each stamp duty change. We construct equally weighted portfolios with sample A and H shares around each event, respectively. In Columns 3 and 4, firm-level returns are examined as $r_{m,i}^A = \gamma_0 + \gamma_1 r_{m,i}^H + \gamma_2 \text{TaxHike}_m^A + C_m + F + \varepsilon_{m,i}$. The dependent variable is return of A-shares over the [+2day,+1day] window around each stamp duty change, while H-share return serves as a control in the regression. Main independent variable *TaxHike* takes two forms: (1) *TaxHike (dummy)* is a binary variable which equals +1 if it's a tax increase, and equals -1 if it is a tax reduction, and (2) *TaxHike (value)* is a continuous measure of the actual increase in stamp tax and its unit is 10 basis point. Firm fixed effects are controlled in some of the specifications. In addition, we control the 12-month base interest rate for China and its difference from Hong Kong's. Robust standard errors adjusted for heteroskedasticity are reported in parentheses. ***, **, and * indicate statistically significant at 1%, 5%, and 10% levels, respectively. Constant terms are omitted in reporting.

Variables	(1)	(2)	(3)	(4)
	$R(A)-R(H)$		$R(A)$	
<i>TaxHike (dummy)</i>	-0.0074** (0.0017)		-0.0142 (0.0093)	
<i>TaxHike (value)</i>		-0.0036 (0.0018)		-0.0062** (0.0025)
<i>R(H)</i>			0.6956*** (0.0993)	0.6860*** (0.0995)
<i>Rf_cn</i>	-0.1139 (0.4966)	0.1015 (0.3395)	0.2770 (0.5899)	0.5803 (0.5979)
$\Delta Rf (cn-hk)$	2.6872*** (0.3936)	2.4955*** (0.2677)	0.9776 (0.6172)	0.6762 (0.5979)
Observations	7	7	223	223
Firm fixed effects	N/A	N/A	Yes	Yes
Adj. R-square	0.84	0.87	0.42	0.43

Table 4: price response to tax-change announcements.

Table 5

Determinants of stamp duty changes.

This table examines potential endogeneity stamp duty changes. Dependent variable is the monthly change in stamp duty in the A-share market from Apr. 1996 to Dec. 2009. In Panel A, the regressors include lags of A-share market returns, and lags of difference in relative return volatility. In Panel B, the sample is restricted to the seven months in which the stamp duty changes actually take place plus the three months each before and after each change. In Panel C, additional controls such as AH_Prem , $float(A)/float(H)$, and $T(A)/T(H)$ are included in the regressions. ***, **, and * indicate statistically significant at 1%, 5%, and 10% levels, respectively. Constant terms are included in the regressions, but omitted in reporting.

Panel A: over period of 1996/04-2009/12				
Variables	(1)	(2)	(3)	(4)
$R(MKT)_{t-1}$	1.95*** (0.57)	1.97*** (0.57)	1.79*** (0.58)	1.77*** (0.58)
$R(MKT)_{t-2}$			0.75 (0.57)	0.78 (0.58)
$R(MKT)_{t-3}$			0.87 (0.58)	0.86 (0.59)
$[V(A)-V(H)]_{t-1}$		-0.60 (2.66)		-4.81 (3.37)
$[V(A)-V(H)]_{t-2}$				2.87 (3.70)
$[V(A)-V(H)]_{t-3}$				2.68 (3.32)
Observations	166	166	164	164
Adj. R-squared	0.06	0.06	0.07	0.07
F-test (market return lags jointly = 0)	11.88***		5.38***	5.28***
F-test (volatility lags jointly = 0)		0.05		0.93
Panel B: months around actual event month				
Variables	(1)	(2)	(3)	(4)
$R(MKT)_{t-1}$	5.12*** (1.69)	5.29*** (1.76)	4.54*** (1.80)	4.58*** (1.80)
$R(MKT)_{t-2}$			1.68 (1.62)	1.64 (1.66)
$R(MKT)_{t-3}$			1.44 (1.80)	1.73 (1.85)
$[V(A)-V(H)]_{t-1}$		-3.14 (3.71)		-12.97 (11.48)
$[V(A)-V(H)]_{t-2}$				5.08 (9.84)
$[V(A)-V(H)]_{t-3}$				6.96 (9.10)
Observations	47	47	47	47
Adj. R-squared	0.15	0.13	0.15	0.13
F-test (market return lags jointly = 0)	8.18***		3.74***	3.23***
F-test (volatility lags jointly = 0)		0.10		0.65
Panel C: additional cross-market features				
Variables	(1)	(2)	(3)	(4)
$R(MKT)_{t-1}$	1.89*** (0.59)	1.71*** (0.58)	1.84*** (0.61)	1.86*** (0.63)
$R(MKT)_{t-2}$	0.64 (0.59)	0.74 (0.57)	0.63 (0.59)	0.53 (0.63)
$R(MKT)_{t-3}$	0.80 (0.60)	0.89 (0.59)	0.67 (0.61)	0.49 (0.64)
AH_Prem_{t-1}	0.55 (1.37)			-0.03 (1.74)
AH_Prem_{t-2}	0.43 (2.05)			1.68 (2.05)
AH_Prem_{t-3}	-0.99 (1.37)			-1.59 (1.75)
$[float(A)/float(H)]_{t-1}$		0.10 (0.17)		0.20 (0.22)
$[float(A)/float(H)]_{t-2}$		-0.36 (0.27)		-0.40 (0.35)
$[float(A)/float(H)]_{t-3}$		0.04 (0.17)		0.17 (0.22)
$T(A)/T(H)_{t-1}$			-0.01 (0.01)	-0.01 (0.01)
$T(A)/T(H)_{t-2}$			0.01 (0.01)	0.01 (0.01)
$T(A)/T(H)_{t-3}$			0.01 (0.01)	0.01 (0.01)
Observations	164	164	164	164
Adj. R-square	0.06	0.06	0.07	0.05

Table 5: determinants of stamp-duty changes.

6.5 Table 6: Stamp duties and stock price volatility; Table 7: Alternative specification

Read: Table 6 reports the baseline volatility regressions. The *HighTax* coefficient is negative across the main firm-level specifications. Table 7 uses alternative volatility measures, including log volatility ratios and variance differences.

Economic message: The average effect is that higher transaction taxes reduce relative A-share volatility.

Interpretation: The result is robust to firm fixed effects, event fixed effects, alternative windows, Fama-MacBeth estimation, and alternative dependent variables.

Interpretation: The continuous specification shows that larger tax increases are associated with larger reductions in relative volatility.

Table 10

Portfolio price volatility. Following Jones and Seguin (1997), equally weighted portfolios and daily price volatilities in 240 trading days before and after each stamp duty change are used for regressions. Robust standard errors adjusted for heteroskedasticity and autocorrelation are reported in parentheses. ***, **, and * indicate statistically significant at 1%, 5%, and 10% levels, respectively. Constant terms are omitted in reporting.

Variables	V(A)-V(H)		Log(V(A)/V(H))	
	Full sample (1)	Subsample (2)	Full sample (3)	Subsample (4)
HIGH_TAX	-0.0116*** (0.0016)	-0.2025*** (0.0233)	-0.0110*** (0.0016)	-0.1863*** (0.0226)
Rf_cn	0.3626*** (0.1085)	8.1488*** (1.4954)	0.4584*** (0.1072)	9.5949*** (1.4007)
$\Delta Rf(cn-hk)$	-0.2076*** (0.0484)	-3.7206*** (0.6892)	-0.2157*** (0.0475)	-3.7777*** (0.6729)
Observations	3356	3356	3356	3321
Month fixed effects	Yes	Yes	Yes	Yes
Event fixed effects	Yes	Yes	Yes	Yes
Adj. R-square	0.09	0.21	0.11	0.26

Table 10: portfolio-level volatility estimates.

Effect of stamp duty change (actual values in percentage point).

Instead of a dummy for a high-tax regime, changes in the actual values of the stamp duty (in percentage point) are used. Robust standard errors adjusted for heteroskedasticity, and firm level clustering are reported in parentheses. ***, **, and * indicate statistically significant at 1%, 5%, and 10% levels, respectively. Constant terms are omitted in reporting.

Variables	V(A)-V(H)			
	Full sample	Subsample	Excl. last 3 events	Full sample
Change in stamp duty	-0.0217*** (0.0031)	-0.0155*** (0.0029)	-0.0592*** (0.0034)	-0.0261*** (0.0028)
Rf_cn	0.4109*** (0.1362)	0.6406*** (0.0986)	0.2232*** (0.0810)	0.9046*** (0.0581)
$\Delta Rf(cn-hk)$	-0.2571*** (0.0201)	-0.2456*** (0.0247)	-0.0323 (0.0574)	-0.4797*** (0.0414)
Observations	446	392	158	446
Firm fixed effects	Yes	Yes	Yes	Yes
Event fixed effects	Yes	Yes	Yes	Yes
Estimated over	1 year	1 year	1 year	6 month
Standard error cluster	Firm	Firm	Firm	Firm
Adj. R-square	0.56	0.58	0.83	0.57

Table 11: continuous tax-change estimates.

6.8 Table 12: Effects on share turnover; Figure 4: Institutional investors and turnovers at the stock level

Read: Table 12 shows that higher stamp duties reduce relative A-share turnover. Figure 4 shows the distributional increase in institutional investors and institutional turnover at the stock level between 2000Q4 and 2008Q4.

Economic message: Taxes work partly by reducing trading activity. But the interpretation of that trading reduction depends on whether the marginal trader is noise-based or fundamental-based.

Interpretation: Figure 4 prepares the heterogeneity analysis by documenting that institutional trading becomes relevant at the stock level by the late sample.

Table 12

Effects of transaction costs on share turnover. Daily share turnover is calculated as $\frac{\text{turnover}_{i,t}}{\text{price}_{i,t} \times \text{share}_{i,t}}$ for event m , stock i , at day t , and dependent variable is the log ratio of A-share's daily turnover over that of H-shares. The subsample in Column 2 excludes potential outliers with price(A-share)/price(H-share) above the 90th percentile. ***, **, and * indicate statistically significant at 1%, 5%, and 10% levels, respectively. Constant terms are omitted in reporting.

Variables	Log(T(A)/T(H))			
	Full sample	Subsample	Full sample	Full sample
HIGH_TAX	-0.5463*** (0.0553)	-0.5207*** (0.0571)	-0.4196*** (0.0542)	-0.2384*** (0.0559)
Rf_cn	11.9712*** (5.4937)	30.6697*** (5.3737)	48.3133*** (9.0224)	75.4456*** (16.5534)
$\Delta Rf(cn-hk)$	-4.7914*** (1.8040)	-3.9414*** (1.7150)	-7.2766 (5.8913)	2.4721 (5.9127)
Estimated over	1 year		60 trading days	30 trading days
Observations	446	392	446	446
Firm fixed effects	Yes	Yes	Yes	Yes
Event fixed effects	Yes	Yes	Yes	Yes
Standard error cluster	Firm	Firm	Firm	Firm
Adj. R-square	0.75	0.82	0.73	0.71

Table 12: tax effects on A/H relative turnover.

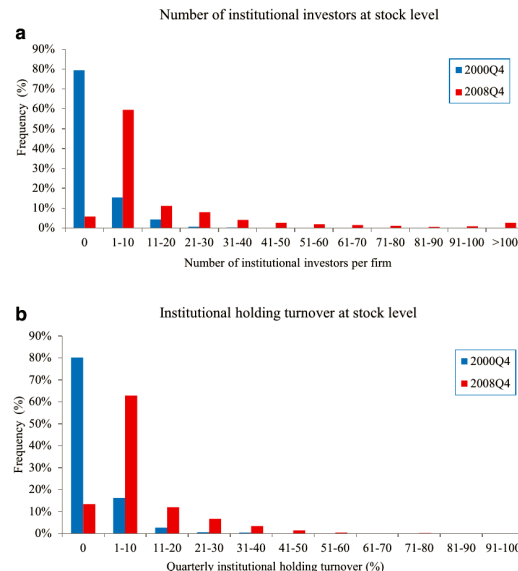


Figure 4: institutional investors and institutional turnover by stock.

6.9 Table 13: Effects over time; Table 14: Separate estimates by event

Read: Table 13 splits the sample into three subperiods. Table 14 reports event-by-event changes in volatility, turnover, and price levels.

Economic message: The negative average effect is strongest early and weakens or reverses later, consistent with the shift toward institutional trading.

Interpretation: Event-level evidence helps show that the pooled estimate masks important heterogeneity across policy events.

Table 13

Effects of stamp duties on price volatility over time. We divide the history of the Chinese stock market into three approximately equally spaced sub-periods: 1990–1997, 1998–2005, and post-2006 period. Two types of regressions are tested: (1) OLS regression; and (2) Fama-MacBeth regression. As the number of cross-listed sample firms in the early stages is small, in the OLS regressions, we follow Cameron et al. (2008) and use wild bootstrap to correct the inferences with a small number of clusters. “***”, “**”, and “*” indicate statistically significant at 1%, 5%, and 10% levels, respectively. Constant terms are included in the regressions, but omitted in the reporting.

Variables	V(A)-V(H)					
	Infant stage (1990–1997)		Toddler stage (1998–2005)		Growth stage (after 2006)	
	OLS	Fama-MacBeth	OLS	Fama-MacBeth	OLS	Fama-MacBeth
HIGH_TAX	-0.0229*** (0.0018)	-0.0145** (0.0065)	-0.0087*** (0.0009)	-0.0043 (0.0093)	0.0011* (0.0006)	0.0180 (0.0207)
Rf_cn	-2.0574* (1.1604)	0.0000 (0.0000)	-0.3971*** (0.0404)	-9.1742 (18.6974)	0.4633*** (0.0570)	-2.5396 (1.6790)
$\Delta Rf(cn-hk)$	1.9194* (0.7919)	-3.8452 (3.9353)	0.2994*** (0.0284)	0.2647 (1.0298)	-0.1515*** (0.0155)	-0.2510 (0.4453)
Observations	348	348	1460	1460	3283	3283
Standard error cluster	Firm	N/A	Firm	N/A	Firm	N/A
Newey-West standard error	N/A	Yes	N/A	Yes	N/A	Yes
Adj. R-square ^a	0.41	0.24	0.22	0.49	0.12	0.27

^a Average R-square is reported for the Fama-MacBeth regression.

Table 14

Separate estimates by event. Changes in price volatility, share turnover, and price levels over 240 trading days before and after each stamp duty change are reported. “***”, “**”, and “*” indicate significance of t-statistics at 1%, 5%, and 10% levels, respectively.

Event date	Stamp duty change direction	Cross-listed firms	$\Delta V(A)-\Delta V(H)$	$\Delta \log V(A)/V(H)$	$\Delta T(A)-\Delta T(H)$	$\Delta R(A)-\Delta R(H)$
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Table 13: effects by market-development period.

12-Jun-98	-	16	0.0060***	0.1020**	0.0077***	-0.0060***
16-Nov-01	-	21	0.0186***	0.5484***	0.0094***	0.0039***
23-Jan-05	-	29	0.0109***	0.4507***	0.0034***	0.0011***
30-May-07	+	42	-0.0022*	-0.1229***	-0.0013***	-0.0034***
24-Apr-08	-	51	-0.0066***	-0.1679***	0.0013*	0.0012***
19-Sep-08	-	51	-0.0115***	-0.3010***	0.0013**	0.0005

i now depends on the relative importance of institutional trading for that stock in the relevant market and time period. The price volatility is still assumed to be the sum of the two components. Importantly, institutional trading affects the price volatility only through its effect on the non-fundamental component.

Similarly, the price volatility in the H market can be expressed as:

we do not need to know the extent of institutional trading in the H -share market (under the assumption that it does not change in the narrow window of a change in the stamp duty in the A -share market).

It motivates us to consider the following regression specification:

$$\Delta \sigma_{m,LL} = \gamma_0 + \gamma_1 \text{High_Tax}_{m,LL}^A + \gamma_2 \text{O}_{m,LL}^A$$

Table 14: event-by-event estimates.

6.10 Figure 5: Changes in relative volatility by event; Figure 6: Changes in relative turnover by event

Read: Figure 5 plots event-level changes in relative volatility against tax changes. Figure 6 plots changes in relative turnover against tax changes.

Economic message: The event-level scatter shows the main directional patterns without relying only on regression tables.

Interpretation: Turnover almost uniformly falls when the tax rises and rises when the tax falls, while volatility responses vary more across market regimes.

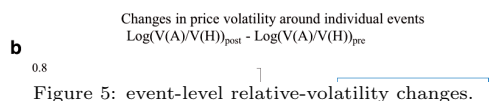
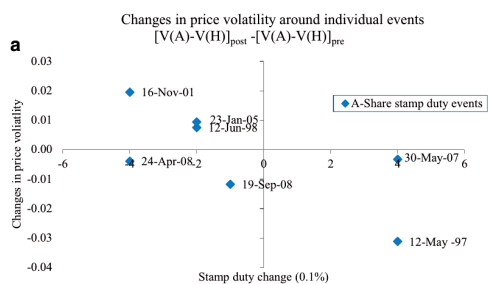


Figure 5: event-level relative-volatility changes.

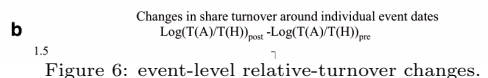
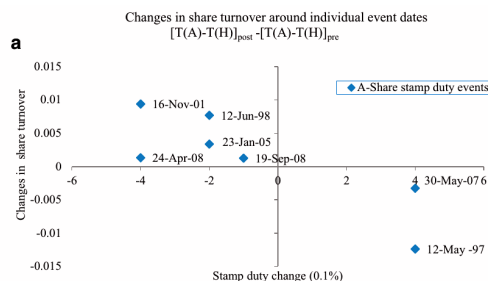


Figure 6: event-level relative-turnover changes.

6.11 Table 15: Institutional shares and price volatility; Table 16: Instrumental variable estimates

Read: Table 15 interacts the high-tax dummy with institutional turnover, institutional ownership, and number of institutional investors. Table 16 instruments institutional trading measures.

Interpretation: The IV estimates sharpen the same conclusion and address measurement error/endogeneity concerns in institutional participation.

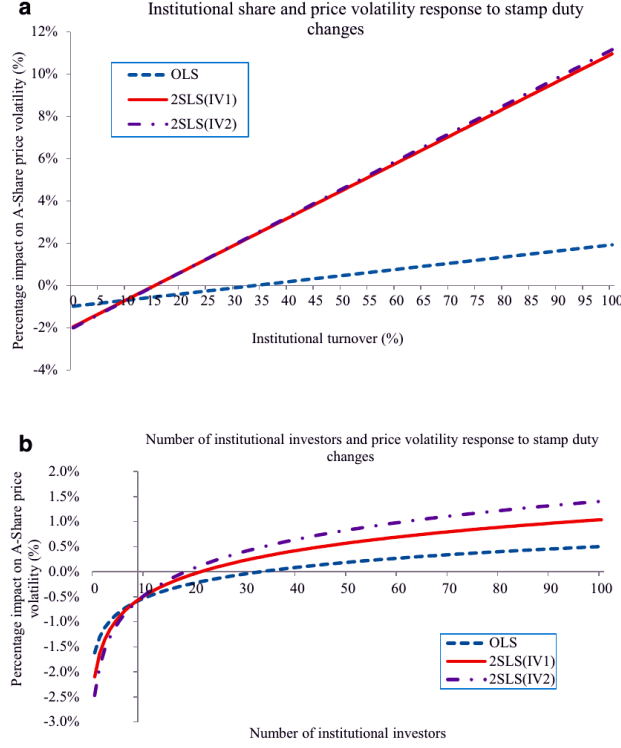
Table 15: institutional participation and volatility response.

Panel A: first stage				
Variables	INST_TURN		INST_SIZE	
	(1)	(2)	(3)	(4)
litig (A)	-0.1386*** (3.0258)	-8.5258*** (2.3287)	-114.1341*** (35.1791)	-123.9504*** (29.0311)
lit(TA)	0.0099*** (0.0041)		0.2484*** (0.0426)	
CSI-300		0.0773*** (0.0176)		1.2412*** (0.1924)
Observations	223	223	223	223
Year fixed effects	Yes	Yes	Yes	Yes
Standard error cluster	Firm	Firm	Firm	Firm
Adj. R-square	0.22	0.25	0.68	0.69
Panel B: second stage				
	V(A)-V(H)			
	IV set 1		IV set 2	
Variables	(1)	(2)	(3)	(4)
HIGH_TAX	-0.0195*** (0.0021)	-0.0210*** (0.0025)	-0.0200*** (0.0022)	-0.0247*** (0.0027)
Inst_turn	-0.0507*** (0.0233)		-0.0640*** (0.0144)	
Inst_turn*HIGH_TAX	0.1291*** (0.0196)		0.1316*** (0.0147)	
Inst_size		-0.0018 (0.0017)		-0.0041*** (0.0009)
Inst_size*HIGH_TAX		0.0068*** (0.0012)		0.0084*** (0.0009)
Rf_cn	0.5732*** (0.0460)	0.5568*** (0.0460)	0.6216*** (0.0460)	0.6209*** (0.0460)

Table 16: instrumental-variable estimates.

Read: Figure 7 plots the estimated tax effect as a function of institutional turnover and the number of institutional investors. The lines cross zero: at low institutional participation, higher taxes reduce volatility; at high participation, higher taxes raise volatility.

Interpretation: The policy implication is conditional: a Tobin tax is not inherently stabilizing or destabilizing. Its effect depends on which traders it deters.



g. 7. Effects of stamp duty on price volatility depends on institutional investor shares. Graphical illustrations of the estimated price volatility response to stamp duty changes as a function of institutional trading are provided below. Results of OLS regression (Columns 1 and 2 of Table 15) and 2SLS (Table 16) are plotted.

Figure 7: Effects of stamp duty on price volatility depends on institutional investor shares. The slope shows the transition from a volatility-reducing tax in retail/noise markets to a volatility-increasing tax in institution-rich markets.

7 Short summary

- The paper studies whether transaction taxes reduce price volatility.
- The setting uses Chinese firms cross-listed as A-shares and H-shares, with one fundamental but two tax regimes.
- A-shares are treated by Mainland stamp-duty changes; H-shares serve as controls.
- On average, higher stamp duties reduce relative A-share volatility.
- Higher stamp duties also reduce turnover.
- Placebo and robustness tests support the baseline result.
- However, the average effect masks important heterogeneity.
- As institutional investors become more important, higher transaction costs raise volatility.
- The same tax can deter noise trading in immature markets and deter informed trading in mature markets.
- The paper reconciles mixed evidence in the Tobin-tax literature by making the effect depend on market maturity.

8 Conclusion

8.1 Core conclusion

Deng, Liu, and Wei show that the effect of a Tobin tax on volatility is not universal. In an immature market dominated by retail and non-fundamental trading, higher transaction costs can reduce volatility. In a more developed market with substantial institutional participation, the same tax can increase volatility by discouraging informed trading and price discovery.

8.2 Economic intuition

A transaction tax changes the composition of traders. If the marginal discouraged trader is a noise trader, prices become less volatile. If the marginal discouraged trader is an informed fundamental trader, prices become more volatile because less information is incorporated into prices.

8.3 Incremental contribution revisited

The paper contributes both a stronger identification strategy and a new conceptual interpretation. Methodologically, the A/H cross-listing design provides a same-firm control group with identical fundamentals. Conceptually, the paper argues that the Tobin-tax debate should be conditional on market maturity and investor composition.

8.4 Policy implications

A transaction tax should not be evaluated as a one-size-fits-all stabilizing tool. It may be useful in young retail-dominated markets, but harmful in institution-rich markets. Policy makers should measure the structure of trading before using taxes to target volatility.

8.5 Limitations

The setting is powerful because of capital controls and A/H segmentation, but the same segmentation also means the results come from a semi-segmented market. The results may not transfer directly to integrated markets where arbitrage is easier or to settings where transaction taxes lead to trading migration.

8.6 Takeaway

The key lesson is that “one fundamental and two taxes” reveals why the literature has mixed results. Taxes can reduce or increase volatility depending on the type of trading they discourage. Market maturity determines the sign.